

Sidhartha Watsa

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EDUCATION

University of California, San Diego (UCSD)

Master of Science, Computer Science (Incoming)

San Diego, CA, USA

Sep 2026 – Present

Indian Institute of Technology (IIT) Kanpur

Bachelor of Technology, Mechanical Engineering (CPI¹: 9.0/10.0)

Double Major, Electrical Engineering (CPI: 9.5/10.0)

Kanpur, India

Jul 2019 – Jun 2023

Jul 2023 – Jun 2024

RESEARCH INTERESTS

Robotic Perception · 3D Vision · Efficient & On-Device AI · Hardware-Software Co-design · Imitation Learning

RESEARCH EXPERIENCE

Samsung Research Institute Bangalore (SRI-B)

Senior Engineer – Next Generation Video Codec at AI Visual Processing Lab

Bangalore, India

Aug 2024 – Present

- Developing core AI-based proposals for next-generation video codec standards beyond [Versatile Video Codec \(VVC\)](#)
- Designing hardware-friendly neural architectures for AI-based In-Loop filters and Intra-Frame Prediction in VVC
- Optimizing on-device AI models for real-time video processing and efficient codec integration on edge devices
- Contributing to open-source codecs: [Advanced Professional Video \(APV\)](#) and [eXtra Essential Video \(XEV\)](#) Codec
- Core intrinsic optimizations for APV and its adoption in [FFMPEG](#), an open-source multimedia framework

Indian Institute of Science (IISc) Bangalore

Project Associate – [Robert Bosch Centre for Cyber-Physical Systems](#), Prof. Ravi Prakash

Bangalore, India

Feb 2024 – Jun 2024

- Investigated imitation learning techniques from imperfect demonstrations across diverse robotic dynamics
- Computed feasibility and optimality scores for each state-action pair in suboptimal human demonstrations
- Trained behavior cloning and inverse reinforcement learning models using refined demonstration subsets
- Simulated model performance on a [KUKA](#) industrial robotic arm using [Gazebo](#) and [Gym](#) environments
- Deployed a complete real-world pipeline for end-effector trajectory planning on a physical robotic arm

Indian Institute of Technology Kanpur

Project Associate – [Intelligent Systems and Controls Lab](#), Prof. Laxmidhar Behera

Kanpur, India

Jul 2021 – Jan 2022

- Developed an unmanned excavator with a 2-DOF robotic arm for remote terrain inspection and data collection
- Integrated gas, microphone, and vision sensors, and implemented fusion algorithms for visual-acoustic analysis
- Designed complete electronics architecture including power management, actuator, and robotic arm control systems
- Formulated decision algorithms for [BIRAC](#)'s brain-controlled wheelchair using Ultracortex Mark IV EEG headset

RESEARCH PROJECTS

End-to-end NNLF Decoder Inference setup in Halide

Samsung Research Bangalore, India

- Investigated whether kernel-level optimization could meet the decoder-side hardware realizability constraint
- Optimized single-core x86 inference of a MAC-bound model using Halide vectorization, cache locality, and scheduling
- Designed modular Halide generators, composing all NN operations into a unified float32 and int16 quantized pipeline
- Delivered 130%² decoder complexity versus 600% for [SADL](#), reduced to 115% with multicore, ready for CFP³ 2027

AI-Based In-loop Filters

Samsung Research Bangalore, India

- Aimed to design a low-complexity neural network (NN) model for suppressing compression and banding artefacts
- Combined handcrafted filters with NNLF, trained on [DIV2K](#), and scaled and blended with the deblocking output
- Delivered state-of-the-art Bjøntegaard Delta Bitrate (BD-BR) gains at ultra-low complexity less than 1kMAC/pix⁴
- Integrated the proposed model within the [VVC Test Model \(VTM\)](#), enabling C++ inference using optimized SADL

AI-Based Intra-Frame Prediction

Samsung Research Bangalore, India

- Examined whether a single neural model could replace seven post-VVC intra-predictors across coding unit sizes
- Designed an intelligent input preprocessing stage to retain strong inter-pixel correlations in reference regions
- Developed a unified neural model featuring multi-scale CNN blocks adaptable to differing spatial resolutions

¹Cumulative Performance Index (equivalent to GPA)

²Decoder time complexity = decoder time with model / base decoder time × 100

³Call for Proposals for post-VVC JVET (Joint Video Experts Team) Meetings

⁴1000 Multiply-ACcumulate operations per pixel

- Integrated the NN-based intra-prediction mode alongside classical predictors in post-VVC, enabling SADL inference

Open Advanced Professional Video Codec (APV)

Samsung Research Bangalore, India

- Contributed core algorithms for APV, an open-source, professional-grade video codec which is perceptually lossless
- Optimized the DCT⁵ transform and quantization steps using platform-specific intrinsic tools (NEON and AVX)
- Popularized APV plugins in FFMPEG, ultrasound edge devices and editing tools like Premiere Pro for NAB demo
- Integrated APV codec end-to-end on Samsung edge devices for Samsung Developer Conference (SDC) demo

IEEE DJI RoboMaster AI Challenge

Prof. Laxmidhar Behera

- Engineered two DJI autonomous robots for intelligent shooting in dynamic arena with advanced AI behaviour
- Customized the YOLO-v5 model for target's armour plate detection with 85% mAP and shooting accuracy of 90%
- Developed motion planning and localization module via visual marker detection combined with rf2o laser odometry
- Utilized DarkNet framework to deploy YOLO-v5 on the outpost camera feeds for global robot and enemy localization

ISRO's Chandrayaan Moon Mapping Challenge

Inter-IIT Tech Meet

- Aim was to train an AI network that produces high-resolution lunar terrain images from medium/low resolutions
- Processed the training dataset (TMC and OHRC images) to obtain ortho-corrected rectangular patches
- Trained and fine-tuned SRGAN cascaded with Shifted Window Transformer (SwinIR) to achieve 16x upscaling
- Achieved SSIM score of 0.71, indicating high similarity between output images and high resolution OHRC images

PUBLICATIONS

- [1] L. Behera, S. Agarwal, T. Sandhan, P. Sharma, A. Kumar, A. Ranjan, S. Watsa, et al., "A cyber-physical system based unmanned ground vehicles for safety inspection and rescue support in an underground mine," *International Journal of Intelligent Unmanned Systems*, vol. 13, no. 1, pp. 92–128, Feb. 2025.
- [2] S. Chikoti, Y. Seelam, A. Jain, N. Mehta, S. Watsa, et al., "Towards autonomous shooting rover via situation aware visual-perception and dynamic action execution," in *2024 IEEE/SICE International Symposium on System Integration (SII)*, Ha Long, Vietnam, 2024, pp. 1113–1118.

PATENTS

Patent Appl. Nos. : 202541023751, 202541024348

- **Inverted-L-Scan Method for AI-based Intra-frame prediction in video codec:** Formulated a pre- and post-processing method utilising an inverted L-scan order to prepare neighbouring pixels for a unified AI-based intra prediction model to preserve inter-pixel correlation in next gen codecs, resulting in enhanced coding efficiency
- **Methods and system for AI-based pixel-block prediction for video codec:** Designed a unified, low-complex neural network architecture for intra prediction with multi-scale CNN block, which leverages neighbouring reference lines to predict coding units (CUs) and is optimised for efficient hardware implementation

HONORS AND ACHIEVEMENTS

- Secured **third position** globally in the IEEE ICRA DJI RoboMaster AI Challenge 2022 among 40+ teams
- Received the Academic Excellence Award (2020–21, 2022–23) **twice** for consistent top **10%** academic standing

TEACHING EXPERIENCE

Indian Institute of Technology (IIT) Mandi

Mandi, Himachal Pradesh, India

Invited by Prof. Laxmidhar Behera, Director, IIT Mandi

May 2022 – Jul 2022

- Designed an institute-wide compulsory interdisciplinary course, "Foundations of Design Practicum", at IIT Mandi
- Conducted a 30-day workshop to train teaching assistants on building a robot from scratch for the course

Samsung Research Institute Bangalore

Bangalore, India

Invited by Learning and Development Team

Oct 2025 – Dec 2025

- Delivered interactive sessions on the fundamentals of device drivers using Linux kernel programming on RPi
- Felicitated with Talent Transformer's Award, 2025 for exceptional contributions in teaching

RELEVANT COURSEWORK

Image processing, Deep learning specialisation, Embedded and Cyber-Physical Systems, Robot Motion Planning

RELEVANT SKILLS

C++, C, Python, Halide, Java, Embedded C, On-Device AI, Android/iOS Programming, OpenCV, OpenAPV, Pytorch, Tensorflow, Robot Operating System, Agentic AI, NEON, AVX, FFMPEG, Gazebo, SADL, DarkNet

⁵Discrete Cosine Transform